Effect of Floating-Body and Stress Bias on NBTI and HCI on 65-nm SOI pMOSFETs

Rahul Mishra, Dimitris E. Ioannou, Souvick Mitra, and Robert Gauthier

***Abstract*—Grounded-body (GB) core-logic/high-speed (HS) and input/output (I/O) silicon-on-insulator pMOSFETs from 65-nm technology are shown to degrade more than floating-body (FB) devices under negative bias temperature instability (NBTI) stress. However, in both cases, worst case degradation occurs when stressed under equal gate and drain voltages ($V_g = V_d$), whereby degradation is simultaneously induced by both NBTI and hot carrier injection (HCI) simultaneously ("concurrent HCI-NBTI"), the relative importance of each mechanism depending on the type of device and the bias level. The degradation of I/O pMOSFETs stressed under $V_g = V_d$ at room temperature shows predominantly NBTI-like behavior at higher stress voltages, whereas it shows concurrent HCI-NBTI behavior at lower stress voltages. By contrast, the degradation of HS pMOSFETs stressed under $V_g = V_d$ shows concurrent HCI-NBTI behavior over the entire stress bias range. In both cases, FB devices degrade more than GB devices for higher stress voltage values, but the FB effects weaken and the degradations become comparable for lower stress bias.**

***Index Terms*—Concurrent HCI-NBTI, hot carrier injection (HCI), negative bias temperature instability (NBTI), silicon-on-insulator (SOI).**

I. Introduction

WITH THE scaling of gate oxide thickness and channel length, pMOSFET hot carrier stress (HCS)-induced degradation departs from the electron trapping mechanism, at low gate voltage V_g/high drain voltage V_d stress [1], to the hot-hole-induced degradation, at $V_g = V_d$ stress [2], [3]. For sub-0.25-μm pMOSFETs, it was concluded [2], [3] that the generation of interface states by hot-holes at $V_g = V_d$ causes the most serious degradation. The situation is more complicated [4] for silicon-on-insulator (SOI) devices as both the worst case stress conditions and the interpretation of degradation are influenced by intricate interdependences of the stress bias level, floating-body (FB) or grounded body (GB) operation, channel-coupling strength, and the self-heating effect (SHE) and device operation temperature. For SOI pMOSFETs under HCS, it has been shown [5] that, as the oxide thickness scales down and the temperature increases (due to the SHE), negative bias temperature instability (NBTI) [6] is also simultaneously activated with the hot carrier injection (HCI). In this letter, we investigate NBTI-induced degradation and its role on the HCS degradation of two families of partially depleted (PD) SOI pMOSFETs: 1) core logic/high speed (HS) and 2) input/output (I/O). Under NBTI stress with GB, these devices degrade more than under NBTI stress with FB. In general, concurrent HC-NBTI degradation was observed under HCS at $V_g = V_d$, except for I/O devices at high stress voltage where NBTI-like degradation dominated.

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R. Mishra and D. E. Ioannou are with the Department of Electrical and Computer Engineering, George Mason University, Fairfax, VA 22030 USA (e-mail: rmishra@gmu.edu; dioannou@gmu.edu).

S. Mitra and R. Gauthier are with the IBM Systems and Technology Group, Essex Junction, VT 05452 USA (e-mail: smitra@us.ibm.com; rgauthie@us.ibm.com).

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II. Experimental Details

HS and I/O PD-SOI pMOSFETs were studied in this letter based on high-performance 65-nm SOI CMOS technology [7]. Both devices that employed (H-gate) body contacts (BCs), which could be independently biased, and FB devices without BCs were available in each case. For HS devices, power supply voltage $V_{\rm dd} = 1.0$ V, gate oxide thickness $t_{\rm ox} = 1.05$ nm, channel width $W = 3$ μ, and channel length $L = 0.06$ μ, and for I/O devices, $V_{\rm dd} = 1.5$ V, $t_{\rm ox} = 2.35$ nm, $W = 3$ μ, and $L = 0.12$ μ. Several HS devices were subjected to NBTI stress for a range of bias and temperature values, and several HS and I/O devices were subjected to HCS at $V_g = V_d$ for a range of bias voltages [5]. The stress was briefly interrupted to measure the device characteristics and the saturation threshold voltage $V_{\rm tSat}$ in both the forward (FWD) and reverse (REV) modes (drain and source interchanged during measurement) to help probe the degradation mechanism. The saturation threshold voltage $V_{\rm tSat}$ was defined as the gate voltage for which the drain current equals 70 nA $\times$ (W/L) for drain voltage $= V_{\rm dd}$.

III. Results and Discussion

Fig. 1 shows the shift of saturation threshold voltage $\Delta V_{\rm tSat}$ versus the temperature under different NBTI stress bias values for HS pMOSFET devices, where the body of the BC devices was grounded (GB operation) both during stress and during measurement. It is seen in this figure that the BC devices degrade more than the FB devices [8] for the higher stress bias and temperature values. This is due to the fact that, for the FB devices, negative charge tunnels from the gate during stress and is being collected in the FB, which effectively lowers the oxide field, thereby reducing NBTI. The lowering of the electric field in this case is supported by the fact that the (tunneling) gate current is lower in the FB device (Fig. 2). Fig. 3 shows the NBTI degradation behavior for I/O devices where, again, the BC(GB)

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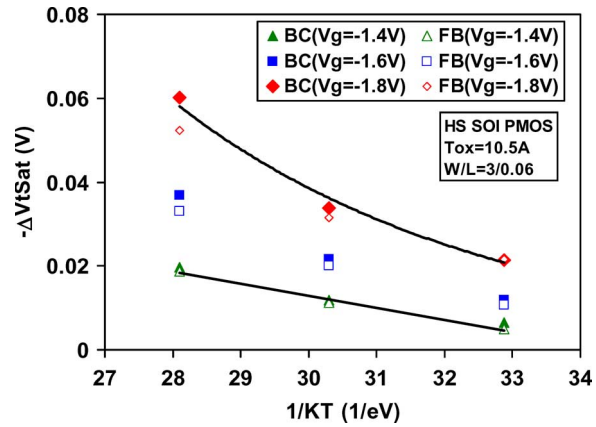

Fig. 1. Saturation threshold voltage shift ΔV_{tSat} versus stress temperature for HS SOI pMOSFET devices. Body-contacted devices with GB [BC(GB)] show more degradation than the counterpart FB devices as the stress bias and temperature value increase.

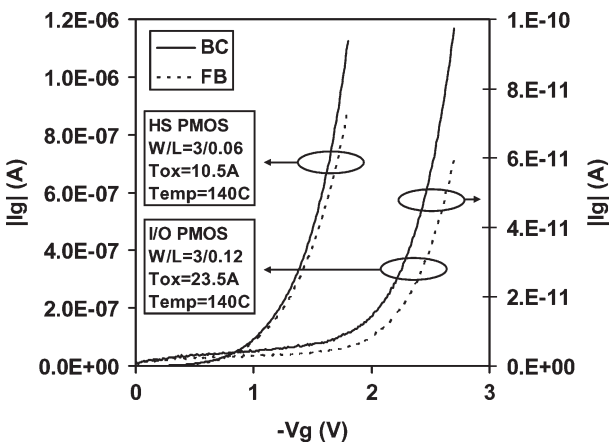

Fig. 2. Gate current comparison of BC(GB) and FB devices for HS and I/O SOI pMOSFETs versus stress temperature. The gate current in the FB device is smaller due to the accumulation of floating charge in the body, and the resulting electric field decreases.

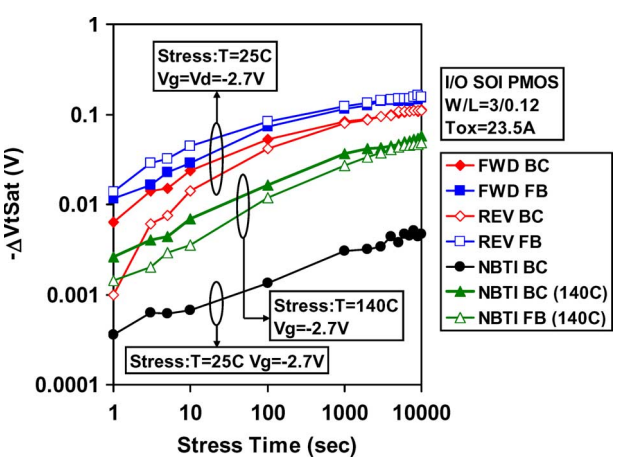

Fig. 3. Saturation threshold voltage shift ΔV_{tSat} versus stress temperature for I/O SOI pMOSFETs under HCS and NBTI stress. The FWD and REV mode curves are on top of each other, demonstrating symmetric degradation along the channel under the present HCS condition. NBTI stress causes higher degradation in the BC(GB) device than in the FB device, whereas the FB device degrades more under HCS.

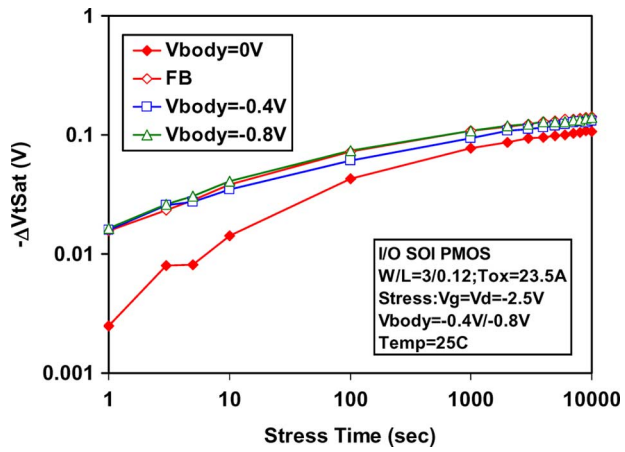

Fig. 4. Saturation threshold voltage shift ΔV_{tSat} versus stress time for BC I/O SOI pMOSFETs for various BC bias (V_{body}) values during HCS. Increasing the body bias increases the pBJT current, leading to FB-like behavior and higher self-heating and, thus, higher degradation than the GB device.

devices degrade more than the FB devices, which is also due to the lowered oxide field in the FB devices, as demonstrated by the lower gate tunneling current (Fig. 2).

Fig. 3 also shows ΔV_{tSat} versus the stress time for both BC(GB) and FB I/O pMOSFET devices under $V_g = V_d = -2.7$ V HCI stress at room temperature. It is seen in this figure that, for both devices, the FWD and REV mode measurements are on top of each other, demonstrating symmetric degradation along the channel. This symmetric shift in V_{tSat} reveals NBTI degradation dominance under the present HCS condition. For comparison, Fig. 3 also shows degradation under NBTI stress at room temperature at $V_g = -2.7$ V (all other terminals grounded). It is seen that the degradation under $V_g = V_d = -2.7$ V (HCI) is much higher than and is parallel to the degradation under $V_g = -2.7$ V (NBTI) stress. This must mean that the self-heating, which is inherent to SOI devices when current flows, enhances NBTI degradation under HCS at $V_g = V_d$ at room temperature. From Fig. 3, it is seen that the FB device degrades more than the BC(GB) device under HCS. We believe that the higher degradation of FB devices under $V_g = V_d = -2.7$ V stress must be due to the parasitic BJT (pBJT), which leads to higher current during stress and, thus, higher self-heating and higher NBTI activation. To verify this, the effect of externally applied body bias on the BC devices during stress was measured, as shown in Fig. 4. It is seen from this figure that the application of body bias V_{body} enhances the degradation and, with $V_{\mathrm{body}} = -0.8$ V, the degradation becomes similar to that of the FB device. This is due to the fact that the application of negative body bias in BC devices turns on the pBJT, which enhances the current and leads to higher self-heating and, hence, higher NBTI, just like that in the FB devices.

Fig. 5 shows ΔV_{tSat} for I/O devices following HCS at $V_g = V_d$ over a stress bias range. It is observed that, at higher stress voltages, the degradation is symmetric, and although there is an increasing difference between the FWD and REV V_{tSat} with decreasing stress voltage, NBTI plays a significant role over the entire bias range.

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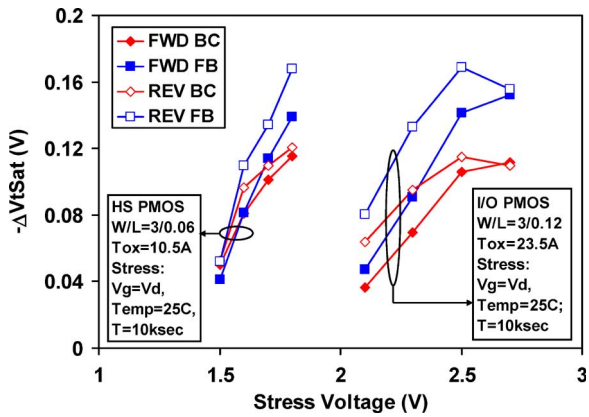

Fig. 5. Saturation threshold voltage shift ΔV_{tSat} versus HCS stress voltage for HS and I/O SOI pMOSFETs. For I/O devices, the degradation is symmetric at higher stress voltages, but the difference between the FWD and REV V_{tSat} increases as the stress voltage decreases. In HS devices, the degradation is asymmetric across the entire stress bias range.

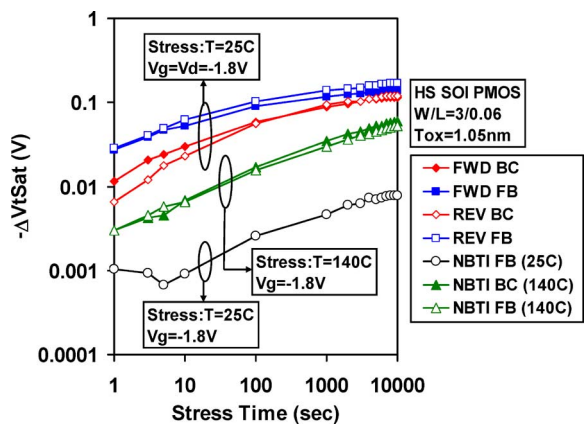

Fig. 6. Saturation threshold voltage shift ΔV_{tSat} versus stress time for HS SOI pMOSFETs under HCS and NBTI stress. NBTI stress causes higher degradation in the BC(GB) device than in the FB device, whereas the FB degrades more under HCS stress.

Fig. 6 shows ΔV_{tSat} for HS devices following HCS at $V_g = V_d = -1.8$ V, as well as that for NBTI stress ($V_g = -1.8$ V). Similar to I/O devices, due to self-heating under $V_g = V_d$ stress, the degradation is higher than and is parallel to the NBTI degradation. However, here, the HCS degradation is not symmetric across the stress bias range, as in the case of the I/O devices previously mentioned (Fig. 5). This demonstrates that, although NBTI activates in HS devices under HCS as well, it is not as dominant as in the case of the I/O devices: In I/O devices, most of the channel remains populated with holes during stress, which cause NBTI-like degradation near the source, and only a small part of the channel near the drain is degraded by hot carriers [9].

IV. Conclusion

Grounded body 65-nm SOI pMOSFETs show higher NBTI degradation than floating body devices due to the lowering of the oxide field in the FB devices caused by gate tunnel current. Both GB and FB devices show worst case degradation under stress at $V_g = V_d$, which is caused by concurrent HCI and NBTI, with NBTI increasingly dominating with increasing bias due to self-heating and for increasing channel length.

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